

# Support Ticket Classification & Prioritization

## 1. Project Objective

The objective of this project is to automatically classify customer support tickets into different categories and predict their priority level using Natural Language Processing (NLP) and Machine Learning. This helps organizations reduce manual effort, improve response times, and efficiently manage customer issues.

## 2. Dataset

Dataset Used: Customer Support Ticket Dataset (Kaggle)

Input Features:

- Ticket Subject
- Ticket Description

Target Variables:

- Ticket Type
- Ticket Priority

## 3. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset. The following analyses were completed:

- Dataset shape and structure
- Missing value analysis
- Duplicate record detection
- Ticket Type distribution
- Ticket Priority distribution
- Ticket text length analysis

## 4. Text Preprocessing

The ticket text was preprocessed using standard NLP techniques including:

- Lowercase conversion
- Removal of punctuation and numbers
- Stopword removal
- Lemmatization
- Combining ticket subject and description

## 5. Feature Engineering

TF-IDF Vectorization was used to convert textual ticket data into numerical feature vectors suitable for machine learning models.

## 6. Machine Learning Models

The following models were trained and evaluated:

- Multinomial Naive Bayes
- Logistic Regression
- Linear Support Vector Machine (Linear SVM)

## 7. Model Evaluation

Models were evaluated using:

- Accuracy

- Precision
- Recall
- F1-Score
- Confusion Matrix

## **8. Results**

The best-performing models were selected based on evaluation metrics. The trained models can automatically predict both the ticket category and its priority.

## **9. Business Impact**

- Reduces manual ticket sorting.
- Improves support response time.
- Prioritizes urgent customer issues.
- Enhances customer satisfaction.
- Increases operational efficiency.

## **10. Conclusion**

This project demonstrates an end-to-end NLP workflow for automating support ticket classification and prioritization. The developed solution can assist support teams in efficiently managing large volumes of customer requests.